

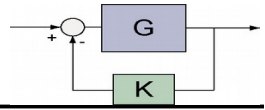
ME 2801

Introduction to Control Systems

W4: Block Diagrams and Feedback



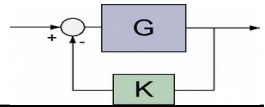
Plan of the Day



Reading Quiz

Block Diagram

- Basic building blocks
- Reduction example, by-hand and with MATLAB
- Feedback example



So Far...

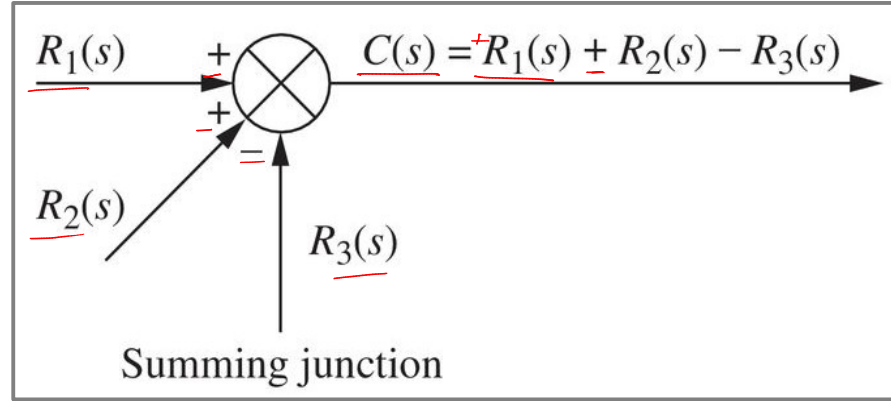
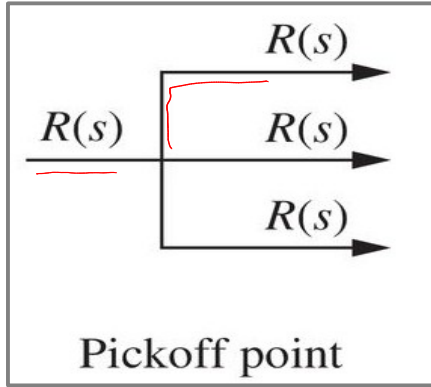
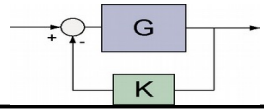
- Laplace transform
- Lumped-parameter models
- First and Second-order response

We can use transfer functions for input-output behavior and s-domain techniques for performance/stability.

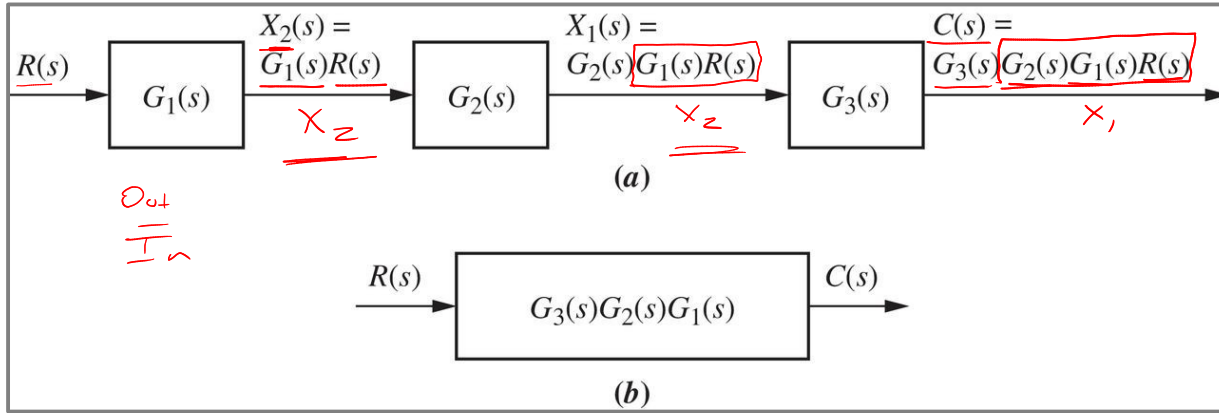
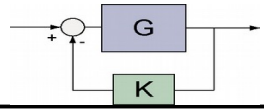
Next...

- Block diagrams allow us to model feedback!
- Predict and design the closed-loop response (performance and stability)

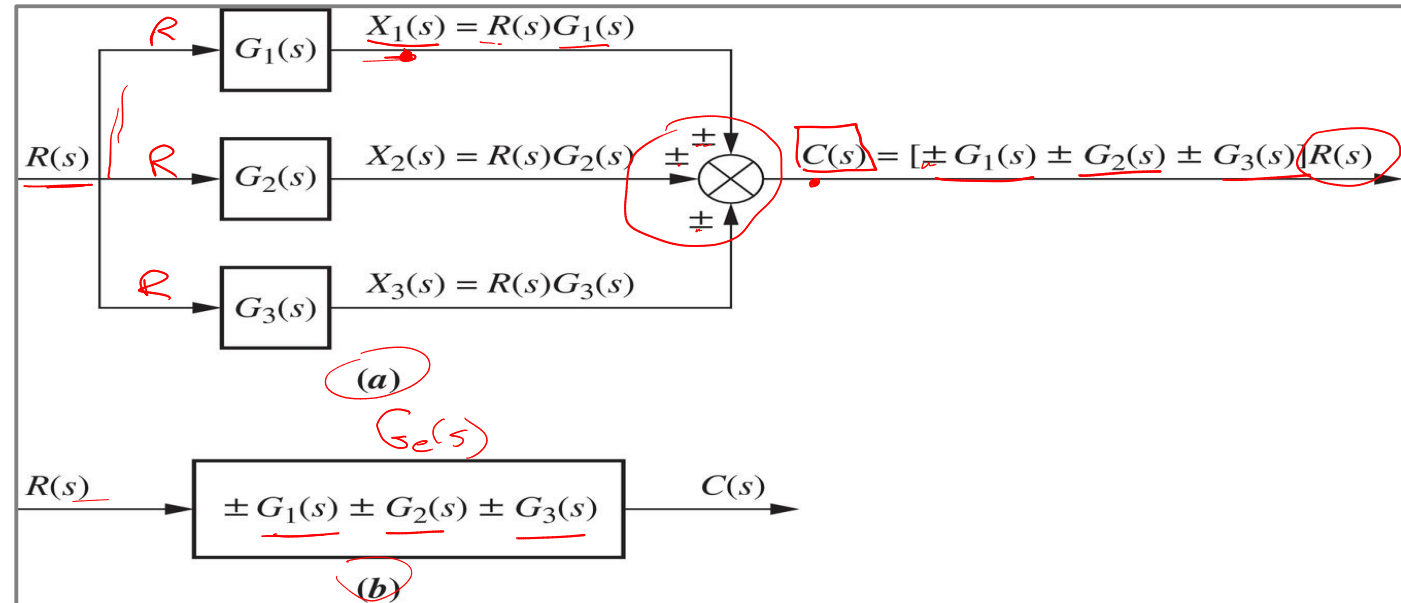
Summing and Pickoff Junctions



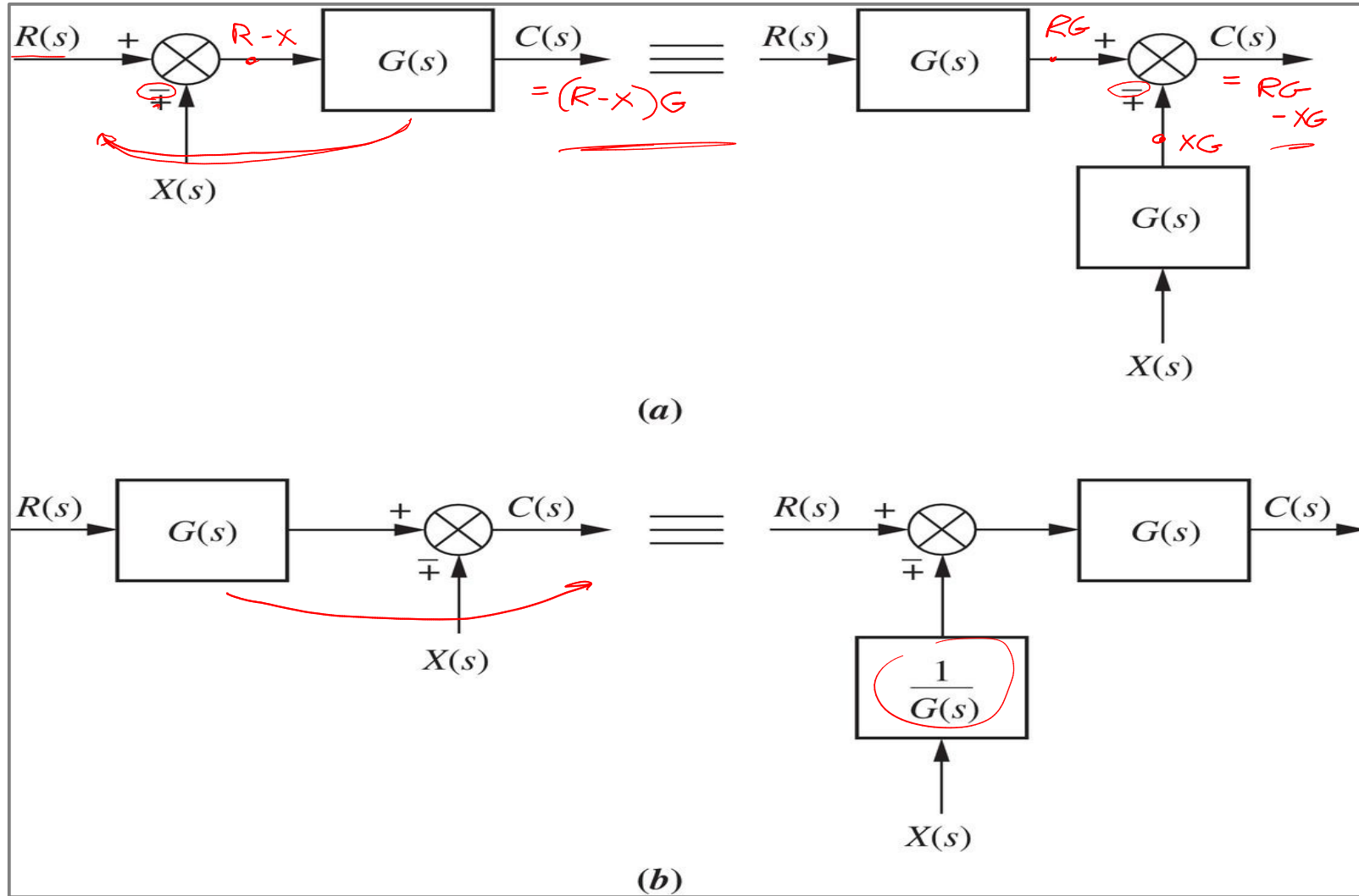
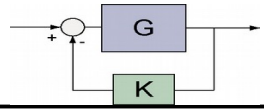
Series (Cascade) and Parallel



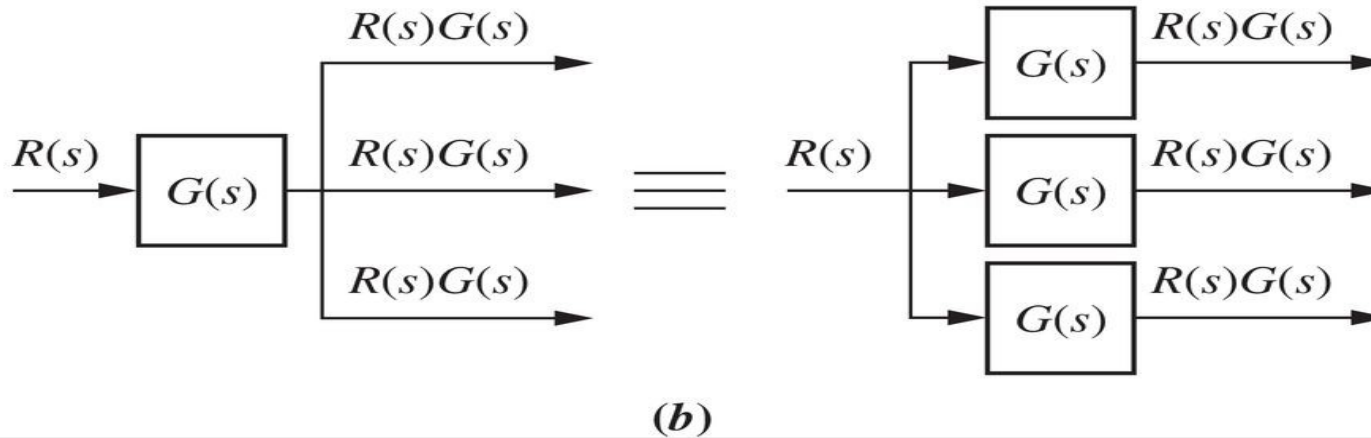
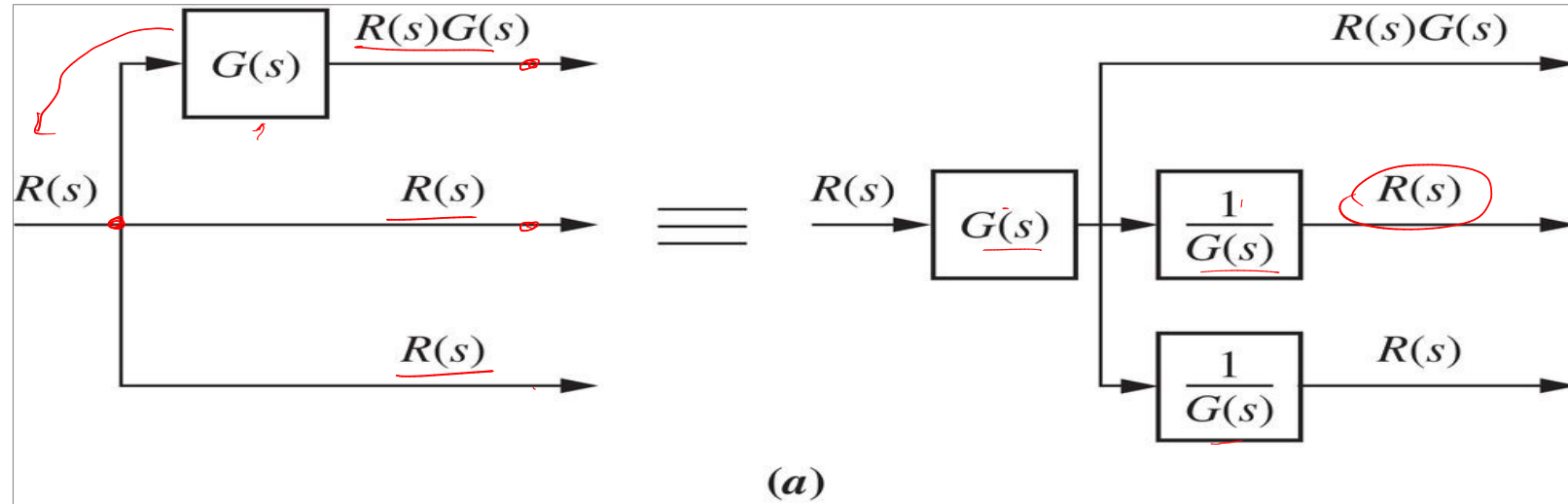
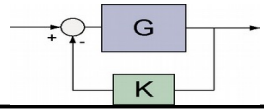
Parallel



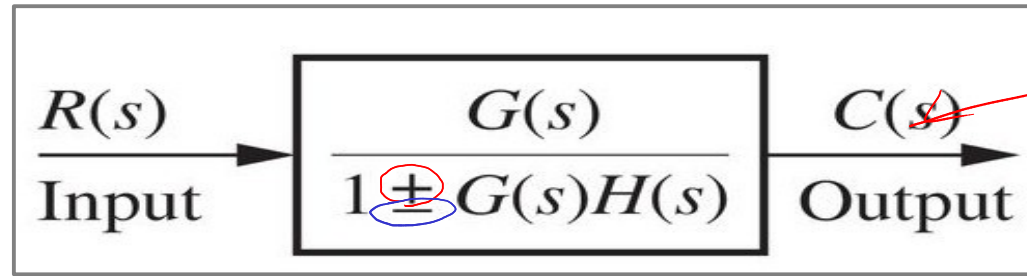
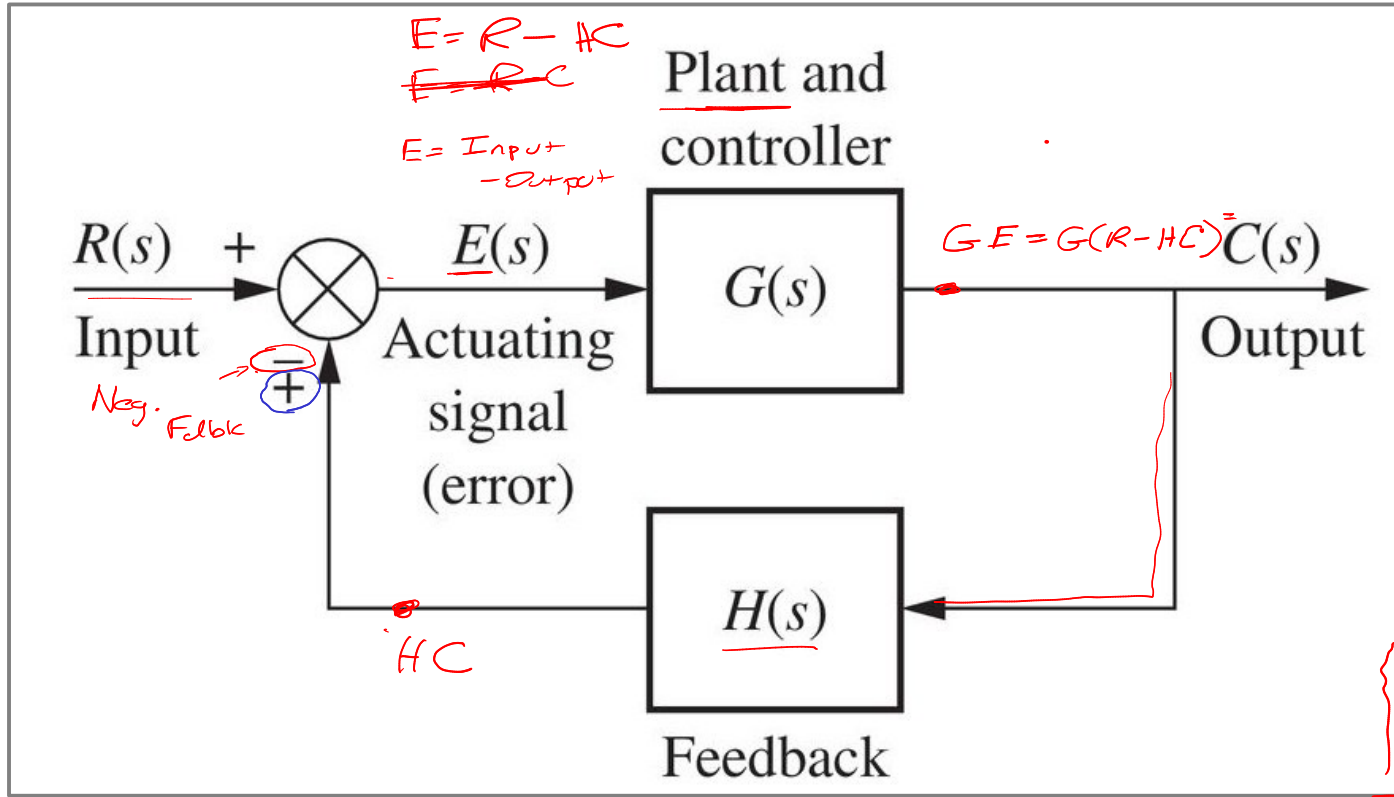
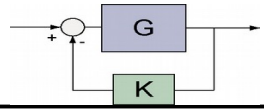
Moving Across Summing Jctn



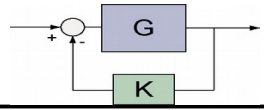
Moving Across Pickoff Jctn



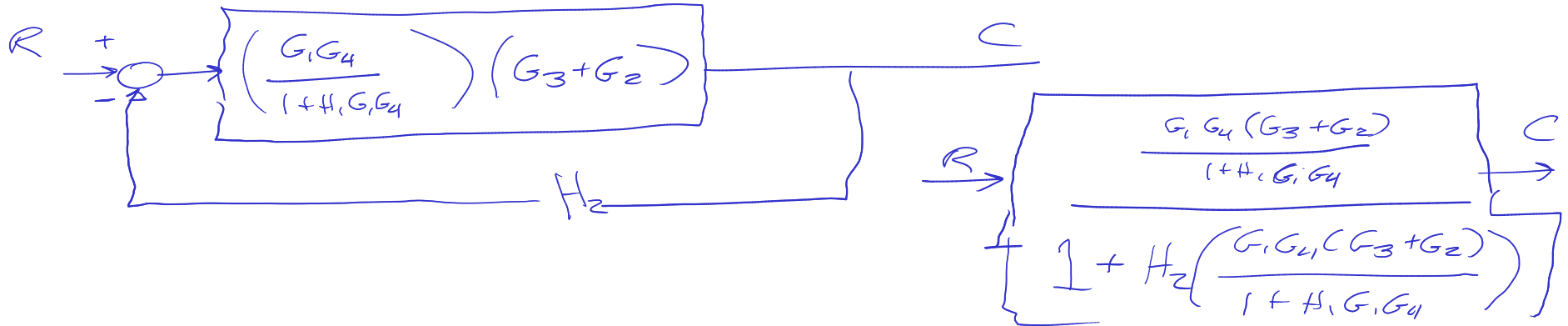
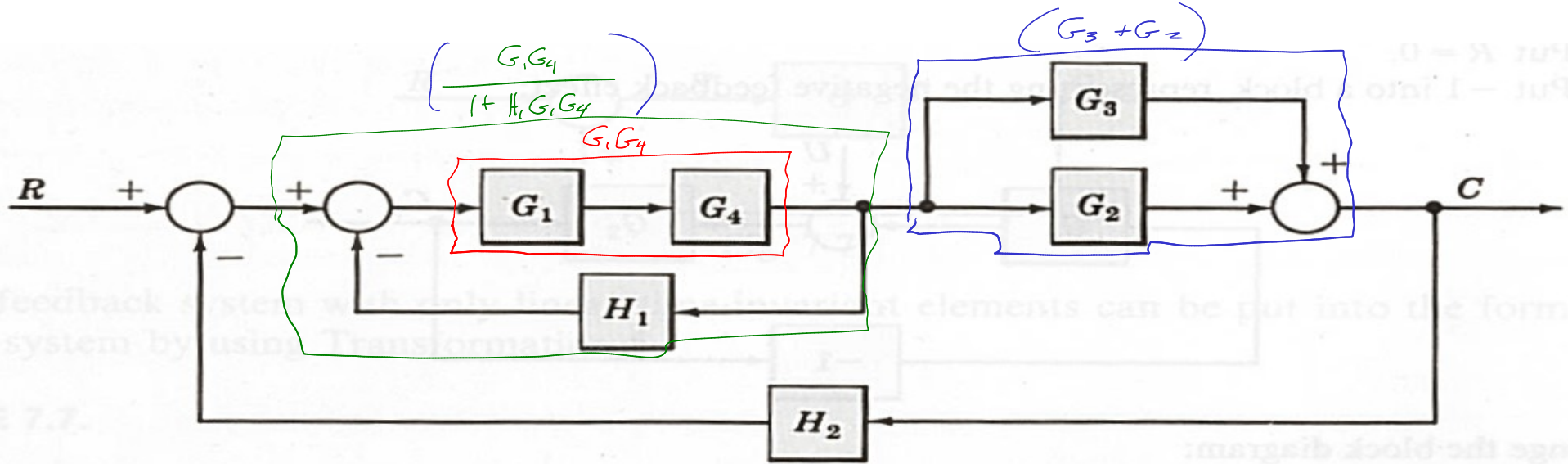
Feedback Form



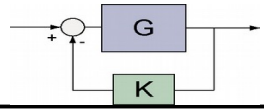
Example



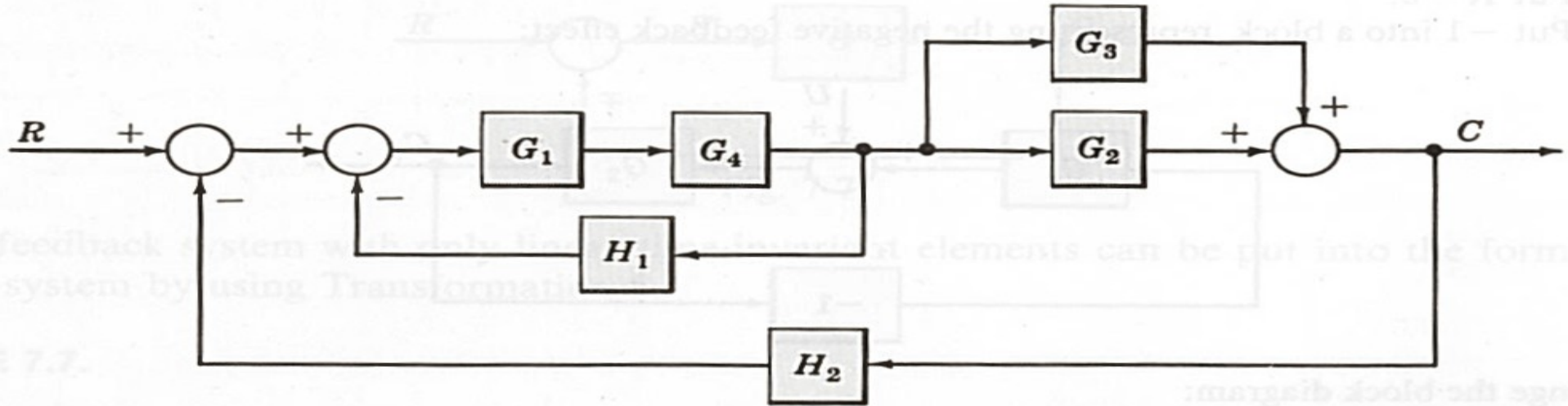
Reduce to single transfer function.



Example: MATLAB



Reduce to single transfer function.



where

$$G_1 = \frac{1}{(s+1)}$$

$$G_3 = \frac{1}{(s+3)}$$

$$H_1 = \frac{1}{s}$$

$$G_2 = \frac{1}{(s+2)}$$

$$G_4 = \frac{1}{(s+4)}$$

$$H_2 = \frac{1}{s^2}$$

$$G1=tf(1,[1 \ 1]);$$

$$G2=tf(1,[1 \ 2]);$$

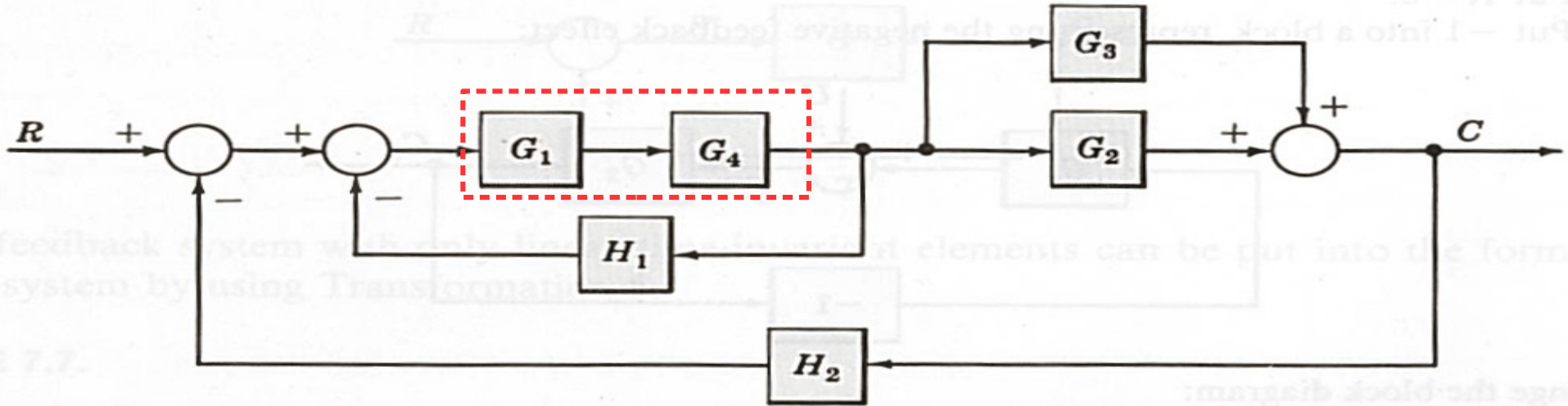
$$G3=tf(1,[1 \ 3]);$$

$$G4=tf(1,[1 \ 4]);$$

$$H1=tf(1,[1 \ 0]);$$

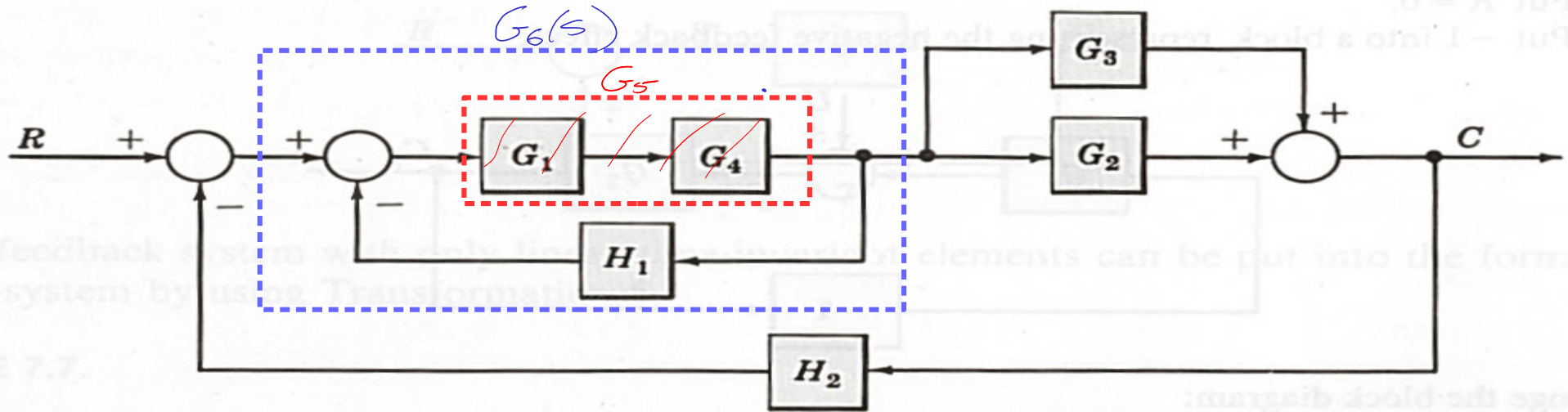
$$H2=tf(1,[1 \ 0 \ 0]);$$

Reduce to single transfer function.



```
G5 = series(G1,G4);
G5 = G1*G4;
```

Reduce to single transfer function.

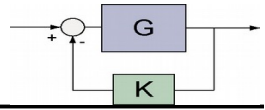


```
G5 = series(G1,G4);
```

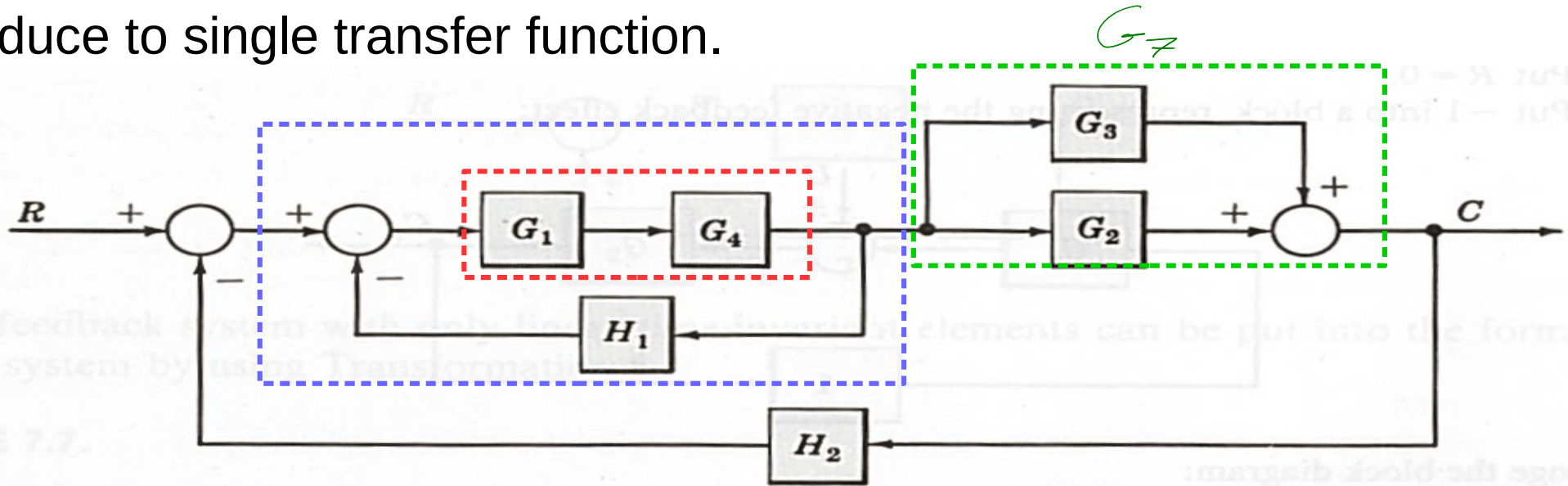
```
G5 = G1*G4;
```

```
G6=feedback(G5,H1);
```

Example: MATLAB



Reduce to single transfer function.



```
G5 = series(G1,G4);
```

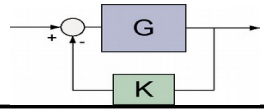
```
G5 = G1*G4;
```

```
G6 = feedback(G5,H1);
```

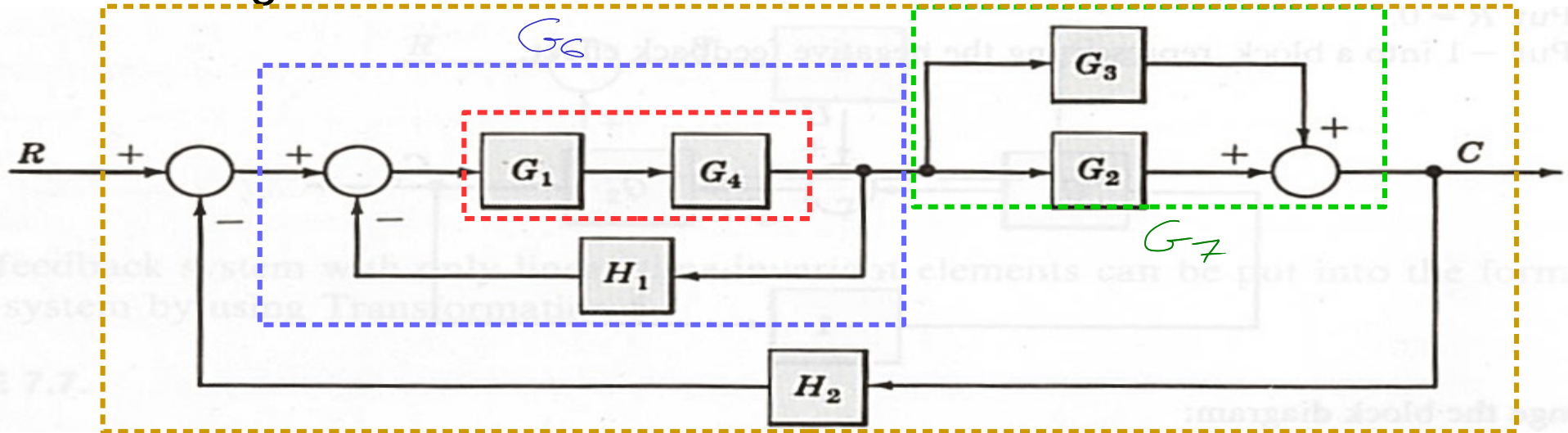
```
G7 = parallel(G3,G2);
```

```
G7 = G3+G2;
```

Example: MATLAB



Reduce to single transfer function.



```
G5 = series(G1,G4);
```

```
G5 = G1*G4;
```

```
G6 = feedback(G5,H1);
```

```
G7 = parallel(G3,G2);
```

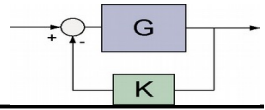
```
G7 = G3+G2;
```

```
G8=feedback(G6*G7,H2);
```

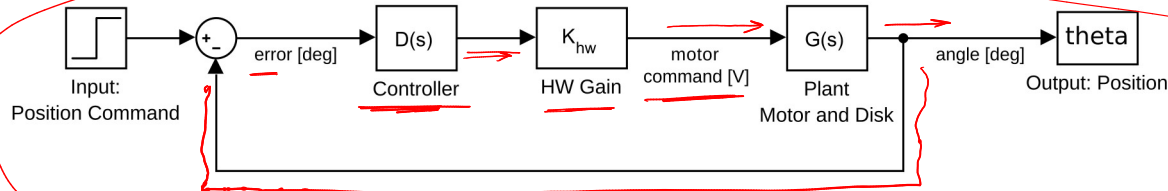
G8 =

$$\frac{2s^4 + 5s^3}{s^7 + 10s^6 + 35s^5 + 51s^4 + 29s^3 + 8s^2 + 5s}$$

ECP Example



Given block diagram

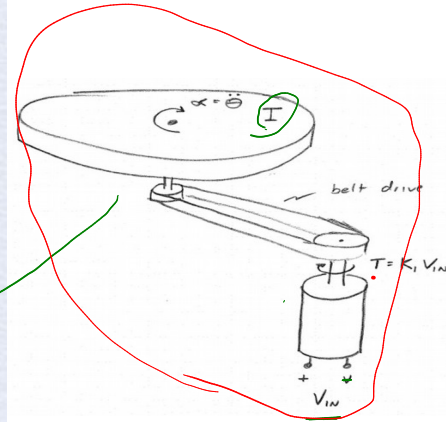
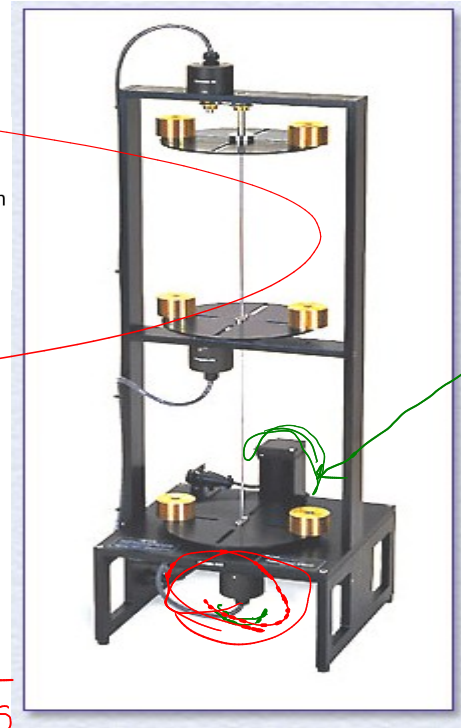


$$G(s) = \frac{\theta}{V_{in}} = \frac{K_m}{s(\tau s + 1)}$$

θ ← angle
 K_m ← Motor Const
 τ ← time const

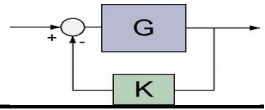
Find closed-loop diagram for proportional control, $D(s) = K_p \frac{V}{deg}$

- Predict the performance (settling time, peak time, %OS, rise time, etc.) as a function of the proportional gain.

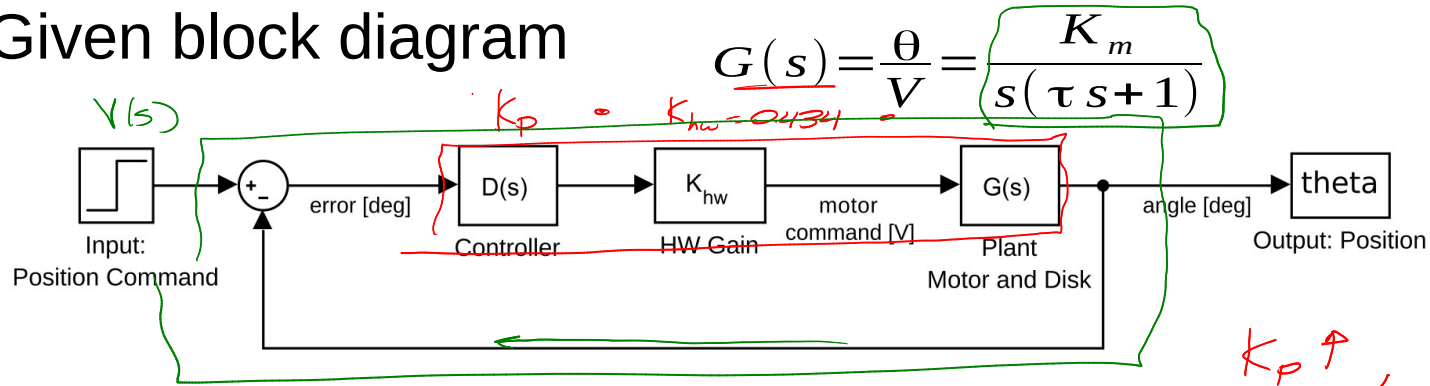


$$T = K_m (V_{in})$$

ECP Example



Given block diagram



$K_p \uparrow, \omega_n \uparrow, \zeta \downarrow, \%OS \uparrow$

Find closed-loop diagram for proportional control

$$D(s) = K_p$$

$2\zeta\omega_n = 1/\tau$
 $\downarrow \uparrow$

- Predict the performance (settling time, peak time, %OS, rise time, etc.) as a function of the proportional gain.

$$G_e(s) = \frac{\theta}{V} = \frac{(K_p K_{hw} K_m) / \tau}{s^2 + (1/\tau)s + (K_p K_{hw} K_m) / \tau}$$

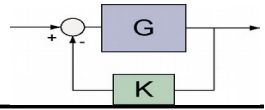
$$\%OS = e^{-(\zeta\pi / \sqrt{1-\zeta^2})} \times 100$$

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$T_s = \frac{4}{\zeta\omega_n}$$

$$T_p = \frac{\pi}{\omega_n \sqrt{1-\zeta^2}}$$

ECP Example



Closed-Loop Transfer Function

$$G_e(s) = \frac{\theta}{V} = \frac{(K_p K_{hw} K_m) / \tau}{s^2 + (1/\tau)s + (K_p K_{hw} K_m) / \tau}$$

$K_p \uparrow, \omega_n \uparrow, \zeta \downarrow, \%OS \uparrow$

% ECP Parameters

```
Khw = 0.434; —  
Km = 25000; —  
tau = 4; —
```

% Proportional Gain

```
Kp = 0.0001; —
```

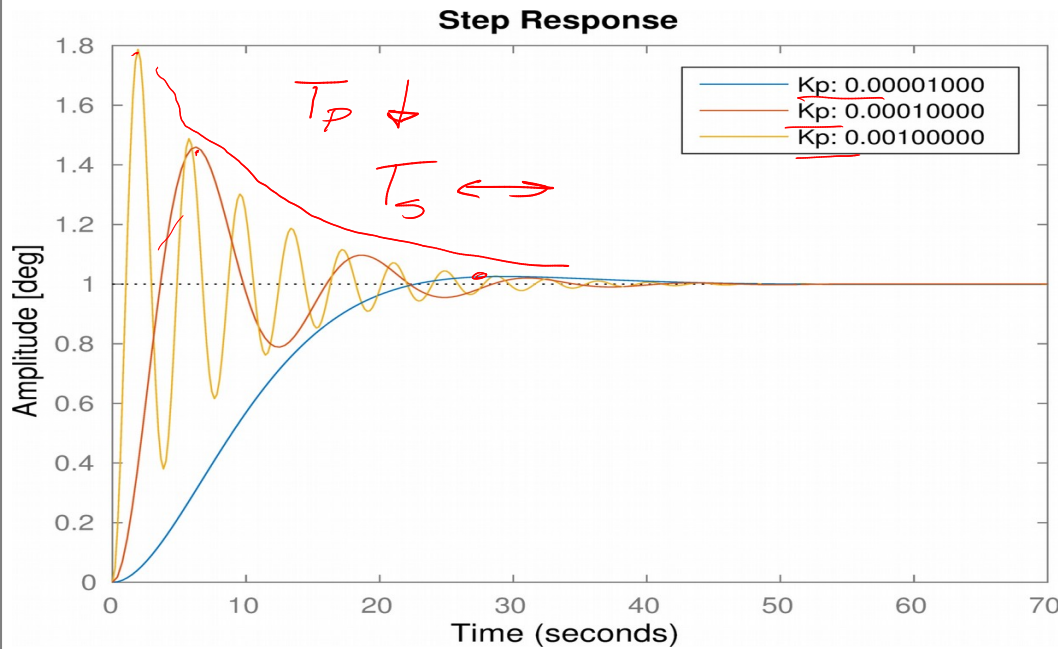
% By Hand

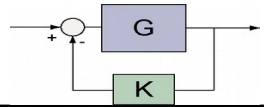
```
Ge == tf(Kp*Km*Khw/tau,...  
[1 1/tau Kp*Km*Khw/tau])
```

% Using MATLAB block diagram algebra

```
G = tf(Km,[tau 1 0]); % ECP plant  
Ge2 == feedback(Kp*Khw*G,1)
```

```
figure(1); clf();  
step(Ge,Ge2)
```





Block Diagrams

- Using visual algebra to manipulate block diagram system descriptions
- Basic forms
 - Summation, pickoff, cascade (series) and parallel
 - Negative feedback

Feedback Design

- Combine
 - Block diagram algebra
 - Transient performance metrics
- Predict the system performance (dynamics) based on simple models of feedback control.